INCH-POUND
MIL-PRF-32192/3
29 July 2005

PERFORMANCE SPECIFICATION

RESISTOR, CHIP, THERMAL (THERMISTOR), NEGATIVE TEMPERATURE COEFFICIENT, STYLE RCTN0404

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RCTN0404 chip thermistors. This style is available in resistance ratios A, B, C, D, H, L, M, N, R, and X. Termination materials G, U, T, C, D, S, and W are available through qualification. Resistance tolerance versus temperature characteristics F, G, J, and K are applicable through the maximum temperature of 125°C.

1.2 Part or Identifying Number (PIN). Chip thermistors covered by this specification are identified by a PIN which consists of the basic number of this specification and a coded dash number. The PIN is in the following form:

M32192	A03G1001GM
Performance specification number	Coded dash number

The coded dash number is derived in accordance with MIL-PRF-32192.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Columbus, ATTN: DSCC-VAT, Post Office Box 3990, Columbus, OH 43218-3990, or emailed to Resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 5905

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2.2 Government documents.

2.2.1 Specifications, standards, or handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-32192 - Resistors, Chip, Thermal (Thermistor), General Specification For.

(Copies of this document are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

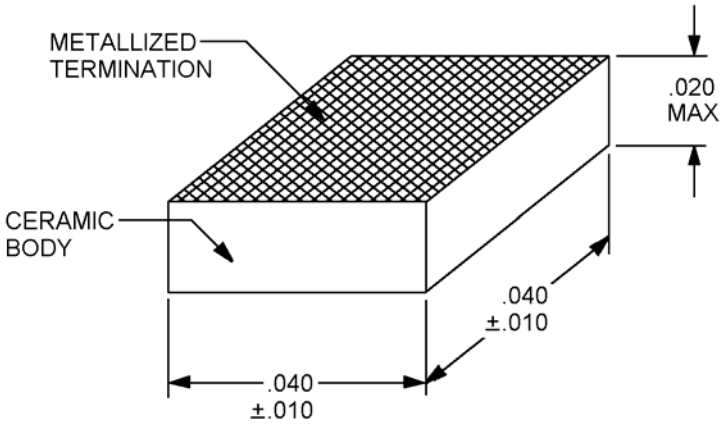
2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-PRF-32192, and as specified herein.

3.2 Interface and physical dimensions. The thermistors shall meet the interface and physical dimensions as specified on figure 1 and herein.

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Inches	mm
.010	.254
.020	.508
.040	1.01

NOTE: Dimensions are in inches. Metric equivalents are for reference only.

FIGURE 1. Style RCTN0404.

3.3 Thermal time constant. The thermal time constant shall be 10 seconds maximum in still air.

3.4 Dissipation constant. The dissipation constant shall be 0.625 milliwatts per degrees Celsius minimum in still air.

3.5 Power rating. The thermistor shall be capable of dissipating a maximum power of 0.0625 watts at 25°C. Thermistors shall be derated in accordance with figure 2.

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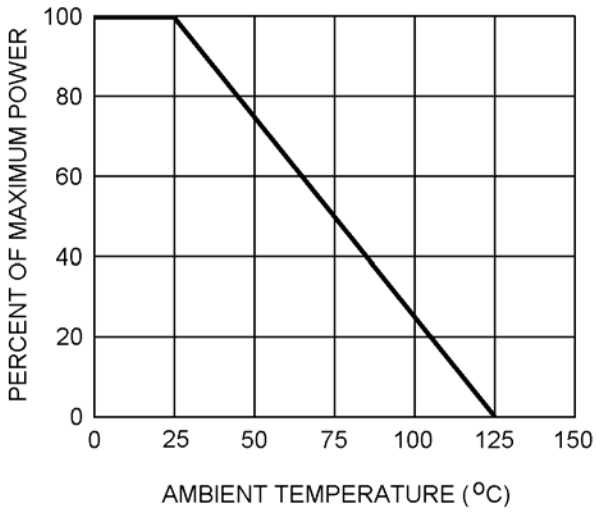


FIGURE 2. Derating curve for high ambient temperatures.

3.6 Resistance.

3.6.1 Resistance at 25°C. The resistance values available are as specified in the table I.

TABLE I. Minimum and maximum resistance values (at 25°C) for each available resistance ratio.

Ratio	Ohms
D (7.1)	15-25
H (13.0)	100-500
L (16.1)	200-500
A (19.8)	1K-3.3K
B (29.4)	3.3K-100K
M (23.3)	10K-30K
N (38.5)	30K-100K
C (48.7)	75K-300K
R (62.5)	500K-2M
X (142.9)	6.8M-20M

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3.6.2 Resistance at temperatures other than 25°C. Resistance at temperatures other than 25°C shall be as specified in table II.

3.6.3 Resistance tolerance. The thermistor specified herein is available in resistance tolerances F (1 percent), G (2 percent), J (5 percent), and K (10 percent) plus the resistance deviation at specified temperature as specified in table II.

TABLE II. Factors for determining resistance at various temperatures.

TEMP °C	Ratio D (7.1)	Ratio H (13.0)	Ratio L (16.1)	Ratio A(19.8)	Ratio B (29.4)	Ratio M(23.3)	Ratio N(38.5)	Ratio C(48.7)	Ratio R(62.5)	Ratio X(142.9)
-55	---	31.9	36.6	54.78	96.40	60.78	---	---	159.0	479.0
-40	6.71	14.4	16.1	22.05	33.66	23.98	40.16	44.03	55.5	128.0
-20	3.47	5.69	6.05	7.422	9.712	7.891	11.03	11.80	14.65	23.38
-10	2.57	3.68	3.89	4.527	5.534	4.754	6.119	6.453	7.61	10.85
0	1.96	2.45	2.57	2.848	3.266	2.949	3.510	3.643	4.09	5.246
15	1.29	1.42	1.43	1.492	1.571	1.513	1.617	1.644	1.72	1.891
25	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
30	0.884	0.854	0.841	0.826	0.806	0.819	0.794	0.787	0.77	0.736
45	0.623	0.537	0.515	0.480	0.437	0.466	0.413	0.398	0.37	0.307
60	0.451	0.350	0.328	0.291	0.249	0.278	0.23	0.212	0.19	0.136
75	0.333	0.238	0.216	0.184	0.148	0.170	0.129	0.118	0.10	0.064
90	0.250	0.163	0.15	0.120	0.092	0.108	0.077	0.068	0.054	0.031
105	0.195	0.117	---	0.081	0.059	0.071	0.048	0.041	0.031	0.016
120	0.157	0.085	---	0.059	0.039	0.048	0.030	0.025	0.019	0.009
125	0.140	0.075	---	0.050	0.034	0.043	0.026	0.022	0.016	0.007

3.7 Short time overload. The maximum allowable change in zero-power resistance as the result of the short time load test shall be as specified in table III.

3.8 Low temperature storage. The maximum allowable change in zero-power resistance as the result of the low temperature storage test shall be as specified in table III.

3.9 High temperature storage. The maximum allowable change in zero-power resistance as the result of the high temperature storage test shall be as specified in table III.

3.10 Thermal shock. The maximum allowable change in zero-power resistance as the result of the thermal shock test shall be as specified in table III.

3.11 Resistance to bonding exposure. The maximum allowable change in zero-power resistance as the result of the resistance to bonding exposure test shall be as specified in table III.

3.12 Moisture resistance. The maximum allowable change in zero-power resistance as the result of the moisture resistance test shall be as specified in table III.

3.13 Life. The maximum allowable change in zero-power resistance as the result of the life test shall be as specified in table III.

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3.14 High temperature exposure. The maximum allowable change in zero-power resistance as the result of the high temperature exposure test shall be as specified in table III.

3.15 Vibration, high frequency. The maximum allowable change in zero-power resistance as the result of the vibration test shall be as specified in table III.

3.16 Shock, specified pulse. The maximum allowable change in zero-power resistance as the result of the shock test shall be as specified in table III.

3.17 Marking (where applicable). The thermistors shall be marked in accordance with MIL-PRF-32192.

TABLE III. Percentage (±) change in zero power resistance by curve.

Test condition	Curve									
	D	H	L	A	B	M	N	C	R	X
Short time overload	2	2	2	2	2	2	2	2	2	2
Low temperature storage	2	2	2	2	2	2	2	2	2	2
High temperature storage	5	4	5	2	2	3	3	2	3	3
Thermal shock	2	3	3	2	2	2	2	2	2	2
Resistance to solder heat	5	5	5	3	3	5	3	3	3	3
Resistance to bonding exposure	5	5	5	3	3	5	3	3	3	3
Moisture resistance	5	5	5	5	5	5	5	5	5	5
Life	5	5	5	5	5	5	5	5	5	5
High temperature exposure 100hr	5	4	5	2	2	3	3	2	3	3
High temperature exposure 1,000hr	25	20	25	3	3	5	5	3	5	5
Vibration	2	2	2	2	2	2	2	2	2	2
Shock	2	2	2	2	2	2	2	2	2	2
Immersion	3	3	3	3	3	3	3	3	3	3

4. VERIFICATION

4.1 Verification. Verification shall be in accordance with MIL-PRF-32192.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military services system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-32192 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Unless otherwise specified (see 2.1), the versions of the individual documents referenced will be those in effect on the date of release of the solicitation.

c. Packaging requirements (see 5.1).

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5905-2014-03)

Review activities:
Army - AR, AT, AV, CR4
Navy - AS, CG, MC, OS
Air Force - 19, 99

Civil agencies:
NASA - NA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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